Feasibility of Automatic Dialog Act Annotation: Adding an Interpersonal Dimension to Clinical Discourse Assessment

SOCKS
The Science of Online Corpora, Knowledge, and Stories

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Story Corps

INTRODUCTION

Individuals with cognitive impairments can exhibit challenges in communication due to the interdependent nature of cognitive processes (i.e., **memory**, attention, and executive function) and communication abilities, which are all essential for maintaining coherence, tracking topics, and **regulating turn-taking in discourse** (McNeil, 2001; Kimbarow, 2021).

is fundamental Assessment of ecologically valid forms of communication, such as conversation, for understanding and treating **cognitive-communicative impairments (CCI)**. Unfortunately, this process is time and labor intensive, and normative data is nonexistent (Togher et al., 2014; Turkstra, Coelho, & Ylvisaker, 2005; VUMC, n.d.)

Dialog acts (DA) represent conversational elements and the relationships between communicative contributions. (Bunt, 2020)

Example of Dialogue Act Segmentation (Želasko, 2021)

A: Yeah, OTHER | Jonathan, uh, you s-, you say it's near and dear to your heart. STATEMENT-NON-OPINION | Buying a car is, uh, something you're contemplating, I take it. DECLARATIVE-YES-NO-QUESTION

B: Uh, fantasizing about. AFFIRMATIVE-NON-YES-ANSWERS

A: Uh. UNINTERPRETABLE

B: Can't do it right now. STATEMENT-NON-OPINION | Our situation is that we're a two car family -- STATEMENT-NON-OPINION

A: Uh-huh. ACKNOWLEDGE-BACKCHANNEL

B: and I have a mini van we bought a few years ago. STATEMENT-NON-OPINION

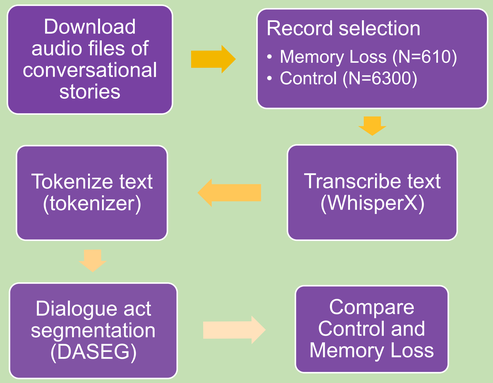
A: Uh. UNINTERPRETABLE

B: It's got a, uh, five year loan, STATEMENT-NON-OPINION | so that's still got a couple years to run -- STATEMENT-NON-OPINION

A: Uh-huh. ACKNOWLEDGE-BACKCHANNEL

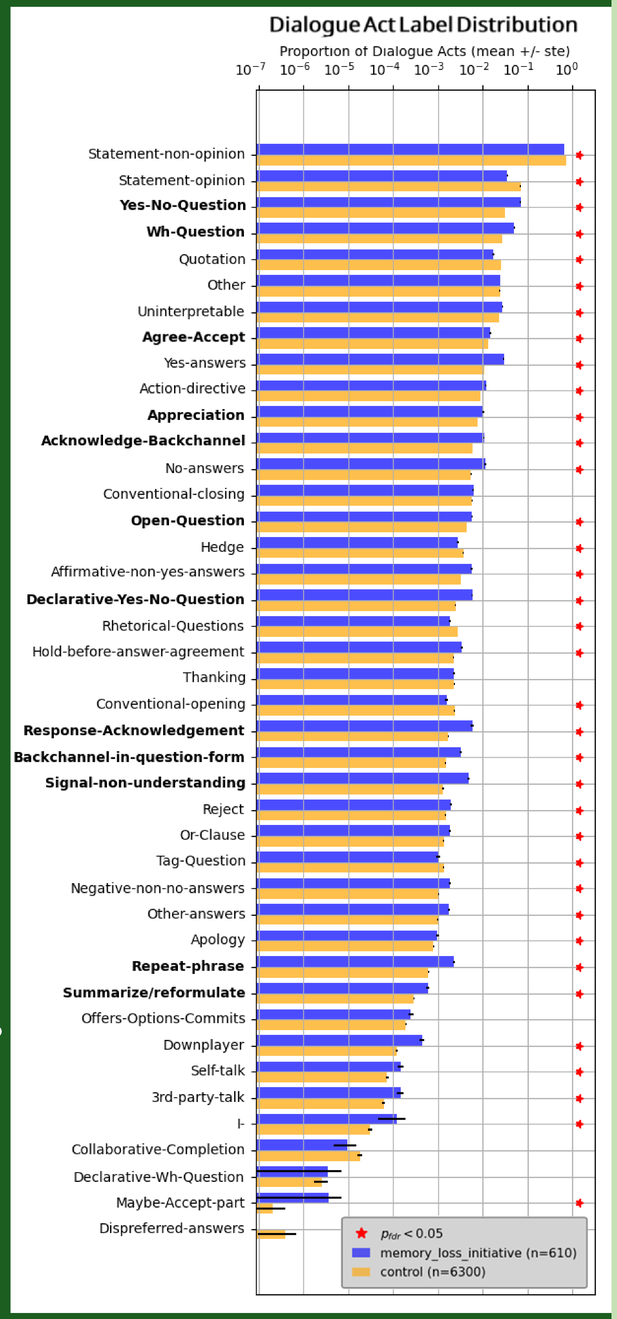
This project uses Želasko et al.'s (2021) longformer-based Dialog Act SEGmentation (DASEG) to automatically segment and label dialogue acts, as a basis for exploring differences in conversational structure among dyads with and without memory loss in the StoryCorps Archive.

METHODS



We applied this pipeline to 6,910 conversations in the StoryCorps Archive, 610 of which were tagged as including someone suffering from memory loss. Conversations are about 40 minutes long. We performed a Mann-Whitney U-test for each dialogue act label, comparing frequencies across control and memory loss stories.

Conversational stories touched by memory loss show signs of increased partner support.



Dialogue Act classes related to partner support are shown in bold.

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Using DASEG

Words Broken Apart By Tokenization (Zimmerman, 2025)

```
A: Yeah, Jonathan, uh, you s-, you say it's near and dear t
o your heart. Buying a car is, uh, something you're contemp
lating, I take it.
B: Uh, fantasizing about.
A: Uh.
B: Can't do it right now. Our situation is that we're a two
car family --
A: Uh-huh.
B: and I have a minivan we bought a few years ago.
```

Želasko et al. trained DASEG to label the first token of the final word within each dialogue act. To use DASEG, we extracted the first token of every word and used the category of the first labeled token as the label for that DA.

RESULTS

- Speed of processing and analysis approximately ten seconds for a single transcript.
- Distribution of labels is comparable to Switchboard.
- High frequency of *Other* labels and reduced counts of *Backchannel* and *Acknowledge* labels (overlapping speech).
- $p<0.05$ for 35 of 42 labels, including categories associated with communication partners' support.

DISCUSSION

We show the feasibility of discourse analysis using automatic dialogue act segmentation and annotation; this could facilitate the use of quantitative social structure metrics in clinical assessments. Memory loss stories included significantly more dialogue acts associated with partner support than control stories. However, the large number of "Other" labels and reduced frequencies of backchannel feedback in comparison with hand-labeled SWBD corpus suggests substantial labeling errors.

Future Work

Quality assurance; Analysis of dialogue act patterns; Descriptive analysis and comparative data creation; Obtain Switchboard data for comparison of category distributions

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Interviews courtesy of StoryCorps, www.storycorps.org

This research is supported by the National Science Foundation under VT EPSCoR Grant No. NSF OIA 2242829. Any opinions, findings, and conclusions or recommendations expressed in this material are those of the author(s)and do not necessarily reflect the views of the National Science Foundation.Computations were performed on the Vermont Advanced Computing Center supported in part by NSF award No. OAC-1827314.

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